Oxidative stability and shelf-life evaluation of selected culinary oils.

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Four out of eight 'healthier' oils-namely, almond oil, avocado oil, hazelnut oil and macadamia nut oil-studied were rich sources of monounsaturated fatty acids like olive oil. Grape seed oil, rice barn oil (marketed recently), toasted sesame oil and walnut oil contained high levels of essential fatty acids. The order of oxidative stability determined by Rancimat measuring of the induction period at four temperatures (90 degrees C, 100 degrees C, 110 degrees C, and 120 degrees C) was found to be macadamia oil > rice bran oil approximately toasted sesame oil > avocado oil > almond oil > hazelnut oil > grape seed oil > walnut oil. High-level monounsaturated fatty acid oils gave a linear relationship between 100 times the reciprocal of the induction period against the total unsaturated fatty acid content obtained as %C18:2 + 0.08 x C18:1 + 2.08 x %C18:3, while the polyunsaturated fatty acid oils gave an exponential relationship. In the case of rice bran and hazelnut oils, shelf-life prediction from the extrapolation of the Arrhenius plots and the Q(10) factors was compared well with that of storage time given by the oil producers. In the cases of the other oils (with an exception of macadamia nut oil), the predicted shelf-lives were significantly lower than that of the storage times; especially, walnut oil (very prone to oxidation) gave 15-20 times lower shelf-life than the best-before storage life.